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DIFFERENCE-IN-DIFFERENCES WITH VARIATION IN TREATMENT TIMING

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post-treatment effect, which is closer to -5. My proposed balance test detects significantly higher per-capita income and male/female sex ratios in reform states, in contrast to joint tests of covariate balance across timing groups, which cannot reject the null of balance. Finally, my results show that much of the sensitivity across specifications comes from changes in weights, or a small number of 2x2 DD's, and need not indicate bias.

I. THE DIFFERENCE-IN-DIFFERENCES DECOMPOSITION THEOREM

When units experience treatment at different times, one cannot estimate equation (1) because the post-period dummy is not defined for control observations. Nearly all work that exploits variation in treatment timing uses the two-way fixed effects model in equation (2) (Cameron and Trivedi 2005 pg. 738). Researchers clearly recognize that differences in *when* units received treatment contribute to identification, but have not been able to describe how these comparisons are made.⁹

The simplest way to illustrate how treatment timing works is to consider a balanced panel dataset with T periods (t) and N cross-sectional units (i) that belong to either an untreated group, U ; an early treatment group, k , which receives a binary treatment at t_k^* ; and a late treatment group, ℓ , which receives the binary treatment at $t_\ell^* > t_k^*$. Figure 1 plots this structure.

Throughout the paper I use “group” or “timing group” to refer to collections of units either treated at the same time or not treated. I refer to units that do not receive treatment as “untreated” rather than “control” units because, while they obviously act as controls, treated units do, too. k will denote an earlier treated group and ℓ will denote a later treated group. Each group's sample share is n_k and the share of time it spends treated is $\bar{D}_k$. I use $\bar{y}_b^{POST(a)}$ to denote the sample mean of y_{it} for units in group b during group a 's post period, $[t_a^*, T]$. $(\bar{y}_b^{PRE(a)})$ is defined similarly.)

⁹ Angrist and Pischke (2015), for example, lay out the canonical DD model in terms of means, but discuss regression DD with timing in general terms only, noting that there is “more than one...experiment” in this setting.

The challenge in this setting has been to articulate how estimates of equation (2) compare the groups and times depicted in figure 1. We do, however, have clear intuition, for 2x2 designs in which one group's treatment status changes and another's does not. We could form several such designs, estimable by equation (1), in the three-group case. Figure 2 plots them.

Panels A and B show that with only one of the two treatment groups, an estimate from equation (2) reduces to the canonical case comparing a treated to an untreated group:

$$\hat{\beta}_{jU}^{2x2} \equiv \left(\bar{y}_j^{POST(j)} - \bar{y}_j^{PRE(j)} \right) - \left(\bar{y}_U^{POST(j)} - \bar{y}_U^{PRE(j)} \right), \quad j = k, \ell. \quad (3)$$

If instead there were no untreated units, the two way fixed effects estimator would be identified only by the differential treatment timing between groups k and ℓ . For this case, panels C and D plot two clear 2x2 DD's based on sub-periods when only one group's treatment status changes. Before t_ℓ^* , the early units act as the treatment group because their treatment status changes, and later units act as controls during their pre-period. We compare outcomes between the window when treatment status varies, $MID(k, \ell)$, and group k 's pre-period, $PRE(k)$:

$$\hat{\beta}_{k\ell}^{2x2,k} \equiv \left(\bar{y}_k^{MID(k,\ell)} - \bar{y}_k^{PRE(k)} \right) - \left(\bar{y}_\ell^{MID(k,\ell)} - \bar{y}_\ell^{PRE(k)} \right) \quad (4)$$

The opposite situation, shown in panel D, arises after t_k^* when the later group changes treatment status but the early group does not. Later units act as the treatment group, early units act as controls, and we compare average outcomes between the periods $POST(\ell)$ and $MID(k, \ell)$:

$$\hat{\beta}_{k\ell}^{2x2,\ell} \equiv \left(\bar{y}_\ell^{POST(\ell)} - \bar{y}_\ell^{MID(k,\ell)} \right) - \left(\bar{y}_k^{POST(\ell)} - \bar{y}_k^{MID(k,\ell)} \right) \quad (5)$$

The already-treated units in group k can serve as controls even though they are treated because treatment status does not change.

My central result is that any two-way fixed effects DD estimator is a weighted average of well-understood 2x2 DD estimators, like those plotted in figure 2. To see why, first assume a

$$\begin{aligned}\hat{\beta}_{k\ell}^{2x2,k} &\equiv \left(\bar{y}_k^{MID(k,\ell)} - \bar{y}_k^{PRE(k)}\right) - \left(\bar{y}_\ell^{MID(k,\ell)} - \bar{y}_\ell^{PRE(k)}\right) \\ \hat{\beta}_{k\ell}^{2x2,\ell} &\equiv \left(\bar{y}_\ell^{POST(\ell)} - \bar{y}_\ell^{MID(k,\ell)}\right) - \left(\bar{y}_k^{POST(\ell)} - \bar{y}_k^{MID(k,\ell)}\right)\end{aligned}$$

the weights are:

$$\begin{aligned}s_{kU} &= \frac{n_k n_U \bar{D}_k (1 - \bar{D}_k)}{\widehat{var}(\tilde{D}_{it})} \\ s_{k\ell} &= \frac{n_k n_\ell (\bar{D}_k - \bar{D}_\ell)(1 - (\bar{D}_k - \bar{D}_\ell))}{\widehat{var}(\tilde{D}_{it})} \\ \mu_{k\ell} &= \frac{1 - \bar{D}_k}{1 - (\bar{D}_k - \bar{D}_\ell)}\end{aligned}$$

and $\sum_{k \neq U} s_{kU} + \sum_{k \neq U} \sum_{\ell > k} s_{k\ell} = 1$.

Proof: See appendix A.

Theorem 1 completely describes the sources of identifying variation in a general DD model and their importance. With K timing groups, one could form $K^2 - K$ “timing-only” estimates that either compare an earlier- to a later-treated timing group ($\hat{\beta}_{k\ell}^{2x2,k}$) or a later- to earlier-treated timing group ($\hat{\beta}_{k\ell}^{2x2,\ell}$). With an untreated group, one could form K 2x2 DD’s that compare each timing group to the untreated group ($\hat{\beta}_{kU}^{2x2}$). Therefore, with K timing groups and one untreated group, the DD estimator comes from K^2 distinct 2x2 DDs.

The weights come both from group sizes, via the n_j ’s, and the treatment variance in each pair.¹¹ With one treated group, the variance of the treatment dummy is $\bar{D}_k(1 - \bar{D}_k)$, and is highest

¹¹ Many other least-squares estimators weight heterogeneity this way. A univariate regression coefficient equals an average of coefficients in mutually exclusive (and demeaned) subsamples weighted by size and the subsample x - variance:

$$\hat{\alpha} = \frac{\sum_i (y_i - \bar{y})(x_i - \bar{x})}{\sum_i (x_i - \bar{x})^2} = \frac{\sum_A (y - \bar{y})(x - \bar{x}) + \sum_B (y - \bar{y})(x - \bar{x})}{\sum_i (x - \bar{x})^2} = \frac{n_A s_{xy}^A + n_B s_{xy}^B}{s_{xx}^2} = \frac{n_A s_{xx}^{2,A}}{s_{xx}^2} \hat{\alpha}_A + \frac{n_B s_{xx}^{2,B}}{s_{xx}^2} \hat{\alpha}_B$$

Similarly, the Wald/IV theorem (Angrist 1988) shows that any IV estimate is a linear combination of Wald estimators that compare two values of the instrument. Gibbons, Serrato, and Urbancic (2018) show a nearly identical weighting formula for one-way fixed effects. Panel data provide another well-known example: a pooled regression coefficients equals a variance-weighted average of two distinct estimators that each use less information: the between estimator for subsample means, and the within estimator for deviations from subsample means.